

Fig. 6: An actual-size, single-side PCB for the robotic arm

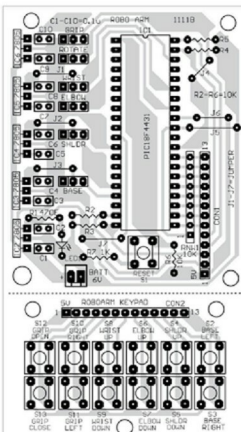


Fig. 7: Component layout for the PCB

tance' represents how far the force is applied from the centre of the shaft.

Fig. 4 shows the torque at various angles. The torque exerted is maximum when applied at an angle of $\theta = 90^\circ$ ($\sin 90^\circ = 1$) and zero when applied at an angle $\theta = 0^\circ$ ($\sin 0^\circ = 0$). To minimise calculations, it is sufficient to assume the worst case of $\theta = 90^\circ$ for servo torque calculations.

Note that when using N-m as the unit of torque, care must be taken to either convert all the loads from kilograms into newtons, or to convert the servo rating in terms of kilograms (1 kilogram = 9.8 newtons).

The torque (T_1) acting on the gripper-servo's shaft is due to the combined weight (W_{G1}) of the gripper mechanism and the rotating gripper's servo. The weight is distributed along the entire length (L_1) but the weight of the load (W_{load}) is acting at L_1 . So T_1 is calculated by the following equation:

$$T_1 = W_{G1} \times L_1/2 + W_{load} \times L_1$$

The weight of the acrylic sheet links is negligible. So W_{L2} and W_{L3} can be ignored. Considering the weights of servos S_1 and S_2 as W_{S1} and W_{S2} , the following equations are obtained for T_2 and T_3 :

$$T_2 = W_{S1} \times L_2 + W_{G1}(L_1/2 + L_2) + W_{load}(L_1 + L_2)$$

$$T_3 = W_{S2} \times L_3 + W_{S1}(L_2 + L_3) + W_{G1}(L_1/2 + L_2 + L_3) + W_{load}(L_1 + L_2 + L_3)$$

The lengths of links L_1 , L_2 , and L_3 need to satisfy the above equations such that the resulting torque value does not exceed the stall torque rating of the corresponding servo motor. The above equations are for the worst-case scenario when the arm is fully stretched. If the torque requirement is satisfied for this condition, normal functioning of the arm will be well within the limits of the servo's torque rating.

Circuit description

Fig. 5 shows the circuit of the robotic arm. PIC18F4431 microcontroller is at

the heart of the controlling circuit. The controller can generate up to six independent PWM signals—one for each servo. The two servos at the shoulder joint share the same PWM signal. The microcontroller has a basic reset circuit and it is powered from a regulated 5V power source obtained through IC2. It is preferable to isolate the microcontroller supply from the servo power source to avoid resetting of the controller due to sudden power surges.

Six PWM output pins of the microcontroller are connected to the six servos. Servos and the control circuitry share a common ground but the servos can be driven directly from the 6V, 4.5Ah lead-acid battery as they draw two to three amperes on full load. Optionally, high-current

voltage regulators may be used.

A single-side PCB layout for the robotic arm (Fig. 5) is shown in Fig. 6 and its component layout in Fig. 7. The keypad can be separated from the PCB by cutting along the dotted lines shown in Figs 6 and 7.

Operation of the arm

This robotic arm has the following articulated anatomy: base, shoulder, elbow, wrist and grip. The top view of the robotic arm is shown in Fig. 8. The operation of different parts of the arm is controlled with six servo motors using twelve keys (S2 through S13) on a keypad. Controlling PWM signals are provided by the microcontroller as discussed below:

Base. The function of the base is to rotate the arm in the azimuth. It is controlled by the PWM output signal from pin 17 of PIC18F4431 (IC1), which drives the base servo motor. Switch S2 is for left rotation and switch S3 for